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INDEX OF REVISIONS

REV.	DESCRIPTION AND/OR REVISED SHEETS
0	ORIGINAL ISSUE
A	WHERE INDICATED
B	TEMPLATE CHANGE
C	REVISED ACCORDING CLARIFICATION NOTICE TO BIDDERS QUESTIONS (item 3 and 6.1)

	REV. 0	REV. A	REV. B	REV. C	REV. D	REV. E	REV. F	REV. G	REV. H
DATE	SEP/13/19	MAY/05/20	JUL/23/20	OCT/26/20					
PROJECT	ESUP	ESUP	ESUP	ESUP					
EXECUTION	SABENÇA	JABLAUSKY	JABLAUSKY	RODBAT					
CHECK	RODBAT	RODBAT	RODBAT	BAVILA					
APPROVAL	CHRISTINO	CHRISTINO	CHRISTINO	CHRISTINO					

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THIS FORM IS PART OF PETROBRAS N-381 REV.J ANNEX A – FIGURE A.1.

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1. INTRODUCTION

1.1 OBJECT

This specification describes and specifies the minimum requirements and criteria, in conformance with relevant regulations and Reference Basic Design contractual documents, for the supply of OILY WATER SEPARATOR PACKAGE (SAO-5330501).

This technical specification will detail the following components:

- Oily water separator package (SAO-5330501)
- Oily water separator package panel (PN-SAO-5330501)

The Oily Water Separator Package shall be provided with all necessary instruments to operate safely, adequately and without interruption in a tropical marine environment.

1.2 DEFINITIONS

CAN: requirements are conditional and indicate a possibility open to the user of the standard.

MAY: indicates a course of action that is permissible within the limits of the standard(a permission).

SHALL: is an absolute requirement, which shall be followed strictly in order to conform with the standard.

SHOULD: is a recommendation. Alternative solutions having the same functionality and quality are acceptable.

PACKAGE: An assembly of equipment supplied interconnected, tested and operating, requiring only the available utilities from the FPSO for full operation.

PACKAGER:
It is defined as the responsible for project, assembly, construction, fabrication, testing and furnishing of the Package.

MANUFACTURER: it is defined as the responsible by fabrication of equipment or components internal to the Package.

HULL SUPPLIER: It is defined as the responsible for all equipment, project, assembly, construction, fabrication, testing, furnishing, installations and services related to Unit Hull.

UNIT: is defined as the FPSO (Floating Production Storage and Offloading), FSO (Floating Storage and Offloading), SS (Semi-Submersible) or Fixed Offshore Unit.

SHIPYARD: the Unit Hull construction and assembly site

OWNER – Petróleo Brasileiro S/A

SAO-5330501: the package name.

1.3 ABBREVIATIONS

FAT Factory Acceptance Tests

FPSO Floating Production Storage and Offloading Unit

QA / QC Quality Assurance / Quality Control

QS Quality Surveillance

1.4 PLANNING

PACKAGER shall prepare an overall project schedule showing design, manufacturing and testing. This schedule shall be submitted with the bid package.

PACKAGER shall prepare a project engineering and manufacturing schedule, and submit the same to HULL SUPPLIER for approval.

The schedule shall be prepared as a bar chart with the main activities shown below:

- DESIGN, showing all issues from basic/outline to certified detail;
- PROCUREMENT, showing raw materials, semi manufactured and bought parts;
- MANUFACTURING, TESTS AND INSPECTIONS, including document output to authorities and HULL SUPPLIER

1.5 SUBCONTRACTORS

PACKAGER shall be responsible for all co-ordination with subcontractors and collections of all details, drawings and data to achieve optimum design and full submission of all documents requested in this specification.

1.6 ALTERNATIVES AND DEVIATIONS

Deviations shall be kept to a minimum. Where PACKAGER considers deviations to this specification and associated reference documents that would result in a more suitable installation, proper modifications may be proposed.

The deviations must be clearly indicated by PACKAGER in a specific item of its Technical Proposal named “Deviation List”, for HULL SUPPLIER approval. Deviations not declared in “Deviation List” shall not be considered and therefore shall be fulfilled by PACKAGER without extra cost for HULL SUPPLIER.

2. NORMATIVE REFERENCES

All equipment shall comply with the requirements of this technical specification, referenced data sheets, documents as stated below and those referred herein.

2.1 INTERNATIONAL CODES, RECOMMENDED PRACTICES AND STANDARDS

The equipment will be designed and manufactured in accordance with the following codes and standards, if not mentioned otherwise.

- DIN German National Standard Code
- ISO International Standard Organization
- EN European Standards
- VDE / IEC German National Electric Standard Codes / International Electric Codes
- BGV German Safety Regulations
- ASME American Society Of Mechanical Engineers
- ANSI American National Standards Institute
- Classification Society defined for the Hull scope.
- Marpol 73/78 - Annex I or most updated
- IMO Resolution MEPC 107 (49) or most updated

2.2 BRAZILIAN CODES AND STANDARDS

- NR 10 Brazilian Federal Government Regulatory Standards (Norma Regulamentadora Nº 10, Segurança em Instalações e Serviços em Eletricidade) - Safety in Electrical Facilities and Services;
- NR 12 Brazilian Federal Government Standards– (Norma Regulamentadora Nº 12, Segurança em Instalações e Serviços em Máquinas e Equipamentos) - Safety on Installation and Services with Machines and Equipment;
- NR 13 Brazilian Federal Government Standards– (Norma Regulamentadora Nº 13, Segurança em Inspeção de Segurança e Operação de Vasos de Pressão, Caldeiras e Tubulações) - Safety on Boilers and Pressure Vessels;
- NR 26 Brazilian Federal Government Standards– (Norma Regulamentadora Nº 26, Sinalização de Segurança) - Safety Signaling;
- NR 37 Brazilian Federal Government Standards– (Norma Regulamentadora Nº 37, Segurança e Saúde em Plataformas de Petróleo) - Safety and Health onboard Offshore Platforms;
- NORMAM-01 – Normas da Autoridade Marítima para Embarcações Empregadas na Navegação em Mar Aberto;

- IBAMA - NOTA TÉCNICA CGPEG/DILIC/IBAMA Nº 01/11 or most updated

2.3 CONFLICTING REQUIREMENTS

In case of conflicting information between this Technical Specification and the referred documents and applicable standards, PACKAGER shall state in its deviation list to HULL SUPPLIER's clarification.

2.4 CLASS APPROVAL AND CERTIFICATION

Certification and approval as required by the above rules is PACKAGER´s responsibility. PACKAGER shall communicate directly and provide all documentation necessary to obtain approvals. HULL SUPPLIER shall be copied on all correspondence. PACKAGER shall obtain approval for all parts of their work before shipment of the equipment to the shipyard.

The Oily water separator package (SAO-5330501) shall be certified by Classification Society defined for the hull scope.

The Oily water separator package (SAO-5330501) shall be designed manufactured and tested in accordance with the latest editions following Rules, Regulations and Standards and reference documents listed on items 2 and 3 of this Technical Specification :

3. REFERENCE DOCUMENTS

Ref Doc Number	Ref Doc Name
~~I-ET-3010.00-0000-91A-P4X-001~~	~~TECHNICAL DOCUMENTATION REQUIREMENTS~~
I-ET-3A36.00-1000-941-PPC-001	METOCEAN DATA
I-ET-3000.00-0000-940-P4X-002	SYMBOLS FOR PRODUCTION UNITS DESIGN
I-ET-3000.00-1200-940-P4X-001	TAGGING PROCEDURE FOR PRODUCTION UNITS DESIGN
I-ET-3010.00-1200-956-P4X-002	GENERAL PAINTING
DR-ENGP-I-1.15	COLOR CODING
I-ET-3010.00-5140-700-P4X-002	SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS

I-ET-3010.00-5140-700-P4X-003	ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS
I-ET-3010.00-1200-800-P4X-002	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS
I-ET-3010.1M-1200-800-P4X-014	AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE
I-ET-3010.1M-1200-200-P4X-002	HULL PIPING SPECIFICATION
I-ET-3010.00-5400-947-P4X-002	SAFETY SIGNALLING
I-ET-3010.1M-1200-300-P4X-002	NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE
I-DE-3010.1M-1200-942-P4X-002	GENERAL ARRANGEMENT
I-DE-3010.1M-5400-94A-P4X-001	AREA CLASSIFICATION – GENERAL
I-RL-3010.1M-1350-960-P4X-009	MOTION ANALYSIS
I-DE-3010.1M-5330-944-P4X-001	BILGE, SLUDGE, BALLAST (ENGINE ROOM) AND GENERAL SERVICE SEA WATER
~~I-ET-3010.1M-1200-310-P4X-001~~	~~API 610 CENTRIFUGAL PUMPS SPECIFICATION~~
~~I-ET-3010.1M-1200-310-P4X-002~~	~~POSITIVE DISPLACEMENT PUMPS SPECIFICATION~~

Note: Reference Documents latest revision shall be considered..

4. QUALITY ENSURANCE AND QUALITY CONTROL

4.1 QA/QC SYSTEM

Engineering, fabrication and manufacturing shall conform to good and sound manufacturing practices. Quality system according to ISO 9001 in relevant extent shall be in place and implemented.

PACKAGER shall submit his Quality Assurance / Quality Control handbook to HULL SUPPLIER for information.

4.2 INSPECTION

HULL SUPPLIER and his designated agents shall have free access to any relevant part of the works, and the right to perform inspections on all aspects of the job covered by this specification and the relevant contract drawings.

HULL SUPPLIER shall have the right to conduct a system audit on the implementation of PACKAGER's quality assurance system. HULL SUPPLIER shall have free access to all relevant documents and records.

OWNER reserves the right to inspect the PACKAGE equipment anytime during fabrication to ensure that material and workmanship are in accordance with this specification.

4.3 QS SCHEDULE

PACKAGER shall propose a quality surveillance schedule with witness and hold points.

5. DESIGN REQUIREMENTS

5.1 DESIGN LIFE

Equipment shall be designed for a 30-year life in a corrosive offshore environment without the need for replacement of any major component due to wear out, corrosion, fatigue, or material failure.

5.2 DESIGN CONDITIONS

PACKAGER shall design the equipment for the full range of operational conditions as specified in this technical specification.

5.3 PERFORMANCE

Oily water separator package (SAO-5330501) shall be designed to guarantee the compliance of the requirements stated in this specification and also on the Technical Data Sheets indicated on the above reference documents (item 3).

All elements of the PACKAGE, including sub orders, shall be of proven design and well within the PACKAGER / MANUFACTURER's actual experience.

5.4 EQUIPMENT LOCATION

Oily water separator package (SAO-5330501) will be installed in the Engine Room, a closed and non-classified compartment.

GENERAL ARRANGEMENT I-DE-3010.1M-1200-942-P4X-002 and AREA CLASSIFICATION PLAN I-DE-3010.1M-5400-94A-P4X-001 shall be used as reference for equipment location.

5.5 SAFETY REQUIREMENTS

Personnel safety protection shall be provided according to Regulatory Standards (NR) issued by Brazilian Federal Government.

Warning signs in Brazilian Portuguese language shall be provided where risk of personnel injury exist.

Rotating equipment outer parts, such as pulleys, couplings, belts and flywheels, shall have rigid protection, manufactured with aluminum ASTM B211 and shall be capable of being easily removed.

In accordance with the requirements of SOLAS II-1, Regulation 3-5, and MSC.1/Circ. 1379, all equipment and material to be supplied by PACKAGER must be “asbestos free”.

PACKAGER shall issue a declaration of “asbestos free” compliance and supporting documentation to substantiate the declaration.

Safety signaling shall be in full compliance with I-ET-3010.00-5400-947-P4X-002 SAFETY SIGNALING.

5.6 NOISE AND VIBRATIONS

Noise and vibrations limits shall conform to I-ET-3010.1M-1200-300-P4X-002 - NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE.

– NOISE AND VIBRATION. Noise data is required for the final proposal and after the Factory Acceptance Tests (FAT).

The maximum sound level at any location 1m from the equipment shall not exceed 85 dB(A). This is applicable to all of the operating conditions for which the equipment is used.

5.7 MOTIONS AND ACCELERATION

All equipment shall be able to withstand when the Unit is subjected to 100-year return period environmental conditions. To operate when the Unit is subjected to 1-year return period environmental conditions defined in I-ET-3A36.00-1000-941-PPC-001 – METOCEAN DATA, at any draft from fully loaded to 20% loaded/ballasted condition, and under inclination (static and dynamic) as specified by Class Society.

The necessary design data and information regarding motion requirements are given in I-RL-3010.1M-1350-960-P4X-009 - Motion Analysis.

The equipment is also to withstand inertial forces during transportation from construction site to operation site (onshore or offshore).

6. PACKAGE SPECIFICATION AND TECHNICAL REQUIREMENTS

6.1 SCOPE OF SUPPLY:

Oily Water Separator Package shall be designed for continuous operation at full load duty, unless otherwise stated in the process data sheets.

The Equipment is intended to perform the bilge water suction from the Engine Room Bilge Water Settling Tank, with a flow rate of 5 m³/h with the maximum oil content of 500 ppm, and discharge this bilge water flow rate to sea overboard with a maximum oil content of 15 ppm, attending IMO MEPC 107 (49) requirements and Marpol Annex I – Reg. 21.

(Note 1) Vendor to inform max. allowable oil content.

The Oily Water Separator Package shall also be automatically shut off in case the level of the Bilge Water Settling Tank is low.

A bilge water oil monitoring system shall be installed in the Oil Water Separator overboard discharge line. This system shall comply with the following requirements:

- One (1) sampling collecting point in the Oil Water Separator overboard discharge;
- One (1) oil cell with range between 0 and 30 ppm of oil;

Oil Water Separator Control Panel PN-SAO-5336501 shall comply with the following requirements:

- Set point of 15 ppm of oil;
- Acoustic and visual alarms and register in control room above 15 ppm of oil;
- Automatic stop of the SAO Bilge Water Pump and close of sea valve after the necessary length of time to block hydraulic hammer effect.

The Oily Water Separator Package shall be PACKAGER's standard design, provided it covers the requirements of this specification. Prototypes are not acceptable.

The complete unit of Oily Water Separator Package and baseplate shall be provided with lugs, etc., to facilitate mechanical handling by means of a single point lift.

The complete package shall be designed, manufactured, tested, inspected and certified to conform to the requirements of this specification and be designed to meet the duty as stipulated on the project data sheets.

PACKAGER shall assume full unit responsibility for the complete package, including drivers and all auxiliaries.

The utility requirements and consumption of the equipment shall be clearly defined by the PACKAGER. This information shall also be included in the quotation.

For overall system schematic, refer to appendix A.

The Oily Water Separator Package (SAO-5330501), including all auxiliary equipment, shall be assembled to the maximum extent possible and pre-checked in PACKAGER's shop, allowing shipment to the conversion yard with minimal fieldwork.

For foreign made equipment, the standard manufacturing parts (couplings, mechanical type seals, anti-friction bearings) shall be purchased from Manufacturers with representative branches located in Brazil, with service parts and maintenance workshops.

The Oily Water Separator Package (SAO-5330501) shall be manufactured, inspected, and verified to comply with all specifications mentioned in Section 2 and the Classification Society regulations.

Dissimilar materials shall be isolated to avoid galvanic corrosion. The assembly shall be designed in such a way to permit:

- Easy accesses for maintenance of all instruments, valves and other components;
- Access for the maintenance and removal of any part of the Unit.

The Unit's weight and dimensions shall be minimized.

Each Unit and respective accessories shall be resistant to the marine environment and shall follow manufacturer standard.
~~API 610 Centrifugal Pumps Specification - I-ET-3010.1M-1200-310-P4X-001; Positive Displacement Pumps Specification - I-ET-3010.1M-1200-310-P4X-002.~~

The PACKAGER shall furnish drawings and procedure for handling and installation of the Units and specify the required devices for the Units maintenance, such as davits, monorails, etc.

Manufactures to supply Data Sheets filled out with all details on the whole of the equipment mounted on the unit. Data Sheets shall contain technical characteristics of equipment such as: type, capacity, rate of flow, pressures, temperatures, materials, diameters of connections, dimensions, weight, air, water and electric consumption, location of center of gravity, electric power rating and such other information as may be deemed necessary.

The package above shall be connected, wired and supplied as a complete unit, ready for installation and operation.

All interconnection piping and valves within the limits of the skid. All necessary ancillaries, instruments and control panel. All other parts or components required for operation of the system.

Additionally, to ensure reliable supplying conditions the Oily Water Separator Package (SAO-5330501) package and associated pumps, control valves, heaters (if applicable), control panels shall be supplied from the same PACKAGER.

7. MANUFACTURING

All materials and equipment supplied by PACKAGER / MANUFACTURER shall be brand new (not overhauled), field proven, free from defects and accepted by Owner and the Classification Society. Materials and equipment shall be manufactured according to internationally recognized standards for the offshore oil drilling and production industries, and shall be in conformance with the Basic Design and Agreement specifications and requirements..

Field proven definition: Systems and equipment shall demonstrate satisfactory operation at least in 3 floating offshore installation units, operating under process conditions (pressure, flow, capacity and similar fluids) for a minimum of 24,000 hours. For rotating equipment, they must demonstrate operation with fluid, flow and discharge pressure similar to the design. Unproven designs or prototypes (including components) without offshore service will not be accepted.

7.1 PAINTING

Painting and coating in accordance with I-ET-3010.00-1200-956-P4X-002 – GENERAL PAINTING and DR-ENGP-I-1.15 COLOR CODING.

The PACKAGER shall submit his painting/coating specification to HULL SUPPLIER for approval.

All components shall be delivered fully painted/coated.

The performed pre-treatment and complete coating shall be in accordance with the paint manufacturer’s data sheets.

7.2 INSPECTION AND TESTING

PACKAGER / MANUFACTURER and their subcontractors shall perform all required inspection and testing in accordance with the referenced design code and/or applicable industry standards. In addition to industry codes and PACKAGER’s standards, PACKAGER / MANUFACTURER and their subcontractors shall comply with the applicable project specifications listed herein, technical data sheets where applicable and with the reference documents mentioned on item 3 of this specification.

PACKAGER / MANUFACTURER shall develop and implement an Inspection and Test Plan (ITP) containing hold points, witness points, schedule of inspection and test, test procedures, tests and events to be witnessed by HULL SUPPLIER´s inspection team.

HULL SUPPLIER representative shall inspect equipment / material at the PACKAGER or MANUFACTURER site and issue the Inspection Release Certificate (IRC).

PACKAGER shall ensure that all the witnessed inspection requirements by the Classification Society are fully accommodated and the due notice requirements are satisfied.

OWNER shall be notified for the inspections events according to the ITP. The notification period for such inspections shall be informed in advance of 30 (thirty) days for foreign PACKAGER / MANUFACTURER and 10 (ten) days for Brazilian PACKAGER / MANUFACTURER.

PACKAGER / MANUFACTURER shall make preliminary test to ensure that all parts of the equipment are operating satisfactory prior to the arrival of HULL SUPPLIER's representative.

If it is found necessary to dismantle any equipment during a test, because of malfunction, the test may then be invalidated, and a full test shall be required after the repair of the fault.

Acceptance of shop tests shall not constitute a waiver of requirements to meet the field tests under specified operating conditions, nor shall inspection relieve PACKAGER of his responsibilities in any way whatsoever.

7.3 FACTORY ACCEPTANCE TEST (FAT)

The following tests shall be included in PACKAGER's / MANUFACTURER's scope:

- Hydrostatic test of all vessels, pipes and pumps;
- Performance test, NPSH test and Mechanical running test of all pumps;
- Electrical continuity checks on all wiring and earthing;
- Functional checks on all instruments and valves;
- Alarms and Equipment Protection Tests;

PACKAGER / MANUFACTURER shall prepare a factory acceptance test procedure (FAT) for the PACKAGE and submit for HULL SUPPLIER approval.

FAT will be witnessed by HULL SUPPLIER representatives. PACKAGER shall advise the HULL SUPPLIER of the test schedule at least ten (10) days for Brazilian PACKAGERS/subcontractors and 30 (thirty) days for foreign subcontractors before the planned test dates. PACKAGER shall invite CLASS surveyor for FAT. HULL SUPPLIER shall invite OWNER according to the same schedule as above.

Acceptance of FAT will not be considered as the final acceptance test of the package.

7.4 COMMISSIONING

PACKAGER / MANUFACTURER shall be required to provide any necessary support for installation and commissioning of the equipment either at a shore based fabrication yard or on the FPSO.

PACKAGER / MANUFACTURER shall provide in the bid the estimated costs for travel and subsistence to and from the site location described in the inquiry. In addition, PACKAGER shall provide labor rates for 12-hours work days onshore and offshore, as well as standby rates for the site location.

PACKAGER / MANUFACTURER shall provide commissioning support as required by the HULL SUPPLIER.

PACKAGER / MANUFACTURER is responsible for assembly supervision of the equipment, including the assembly of components to be delivery loose (for example, some components of the pumps, like stuffing box, etc.).

Final acceptance will be on satisfactory completion of commissioning tests as specified by OWNER.

8. GENERAL REQUIREMENTS

8.1 ELECTRICAL REQUIRMENTS

All electrical equipment installed in hazardous areas (see Area Classification documentation) or installed outdoors and kept on during emergency condition (ESD) shall be certified according to IEC 61892, INMETRO Resolution 179, May 18th 2010 and INMETRO resolution 89, February 23rd 2012.

Electrical equipment and material shall comply with requirements of I-ET-3010.00-5140-700-P4X-002 – SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS.

Concerning electrical system voltages and quantity of feeders for motors, panels and auxiliaries, centrifugal pumps shall be fed according to definitions of I-ET-3010.00-5140-700-P4X-003 – ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS.

8.2 INSTRUMENTATION AND AUTOMATION REQUIREMENTS

The instrumentation and control design shall fulfill the requirements of I-ET-3010.00-1200-800-P4X-002 – AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS.

The minimum requirements for the adequate interfacing of the Main Packages' Automation and Instrumentation System with the UNIT are described on I-ET-3010.1M-1200-800-P4X-014 – AUTOMATION INTERFACE OF PACKAGE UNITS .

8.3 SPARE PARTS, CONSUMABLES AND TOOLS

Spare parts, consumables, and tools must be provided by PACKAGER / MANUFACTURER and included as follows:

a) All equipment's / material's consumable and spare parts recommended by PACKAGER / MANUFACTURER for the construction, testing, commissioning, pre-operation and start-up phases;

b) All spare parts recommended or required by the Classification Society, such spare parts will be delivered together with the relevant equipment;

c) All special tools required for construction, pre-commissioning, commissioning and all levels of maintenance and operation

d) Spare parts list recommended by PACKAGER / MANUFACTURER for two years of operation, including price and delivery time of each part, lay-out and sectional drawings indicating the location of part and TAG/reference identification (part number).

Notes:

All supplied spare parts and tools must be physically identified with the TAG of equipment and its part numbers into boxes to easily storage materials in the warehouse.

When the package is comprised by more than one identical equipment, the spare parts list shall clearly define if the quantity of spare is for one equipment of for all equipment included in the package.

8.4 AVAILABLE ON BOARD

PACKAGER shall consider on the package design the following parameters regarding available on board.

	OPERATING/DESIGN Pressure (barg)	Temperature (C)
Compressed air	9.28/12.26	40
Fresh Water	5.0/7.1	30

- Electric supply: 480 Vac, 60 Hz, 3 (motor up to 150 kW) rated voltage for motor shall be 440 V; 220 Vac, 60 Hz, 1 (for space heaters).

8.5 LAYOUT

PACKAGE layout and arrangement shall be designed to provide sufficient access for ease of operability and maintenance, and to maximize safety.

PACKAGE shall be designed and fabricated such that all equipment and components are located entirely within the skid base perimeter, including all equipment, piping, valves, electrical, instrumentation and controls. The projection of such items beyond the perimeter of the skid base shall be strictly prohibited, unless approved in writing by HULL SUPPLIER.

8.6 NAMEPLATES

PACKAGER / MANUFACTURER Equipment shall have nameplates in Brazilian Portuguese language, made of stainless steel AISI 316L, with 3 mm minimum thickness and fixed by stainless steel (AISI 316L) bolts or fasteners on visible and accessible location. Nameplates shall include at least the following information:

- Owner;

- Petróleo Brasileiro S.A. – OWNER;
- PACKAGER's name;
- Serial number;
- Manufacture Year;
- Main data for design, operation and testing (Power, Pressure, Volume, Temperature, Rotation, Flow rate), where applicable;
- Specific requirements;
- Installation identification;
- Equipment TAG;
- Purchase Order Number;
- Empty Weight;
- Work with Hydrogen – whenever applicable;
- H2S work - whenever applicable;
- Hydrostatic test water requirements – whenever applicable;
- Potential Risk Group Category (for pressure vessels and shells of Shell Heat Exchanger) according to NR13 (identification shall be painted on the body of the vessels and visible at distance of at least 5.0 m). - whenever applicable;
- Design Code;
- Service;

8.7 TAG NUMBERING

Tagging of all instruments, electrical, mechanical and piping items, including valves, shall be carried out. Tag numbers will be supplied by HULL SUPPLIER.

The main items shall have individual tag numbers as dictated by HULL SUPPLIER. The actual tag numbers will be advised to PACKAGER after award.

Tags shall be supplied with the number and description in the Brazilian Portuguese language, unless otherwise stated in the technical data sheets.

Valves, instruments and orifices shall be tagged with the applicable number only. Tag numbers for remaining ancillary equipment shall be given after purchase order placement.

For Instrumentation tagging the ISA –5.1, N-1710 and I-ET-3000.00-1200-940-P4X-001 - Tagging Procedure for Production Units Design shall be followed.

8.8 PRESERVATION, PACKING AND TRANSPORTATION

PACKAGER / MANUFACTURER shall ensure all the conditions and practices of preservation, packing and transportation are fulfilled and following the PACKAGE / Equipment particular and technical characteristics recommendations.

Preservation and packing shall be proper for transportation and storage in a marine environment and protected against moisture and damage during transport, handling and lifting. In any case, suitable preservation and protective measures shall be provided to prevent equipment deterioration prior to entering into service.

All packing shall be clearly marked for shipping, including lifting points, gross weight, dimensions and center of gravity. All sea fastening and temporary supports used on the equipment for shipment shall be clearly identified.

PACKAGER / MANUFACTURER shall ensure that all loose valves, tubes and instruments are supplied with plastic caps. Contractor shall also ensure that all electric panels and motors will be supplied with Volatile Corrosion Inhibitor (VCI) impregnated plastic protection or similar, and external plug for space heater connection.

PACKAGER / MANUFACTURER shall provide clear and comprehensive instructions on the exterior of all packages advising the necessary warning notices for unpacking, handling and installing the equipment on arrival at destination.

The equipment shall be thoroughly cleaned internally and be free of all loose foreign materials.

The preparation shall make the equipment suitable for 12 months outdoor storage in a coastal tropical climate from the time of Shipment.

If there is a risk of damage to valves and other appurtenances during transportation, they shall be disconnected and tagged. All components shall then be securely packed as above.

Spare parts and tools to be packed separately and clearly marked “Spare Parts” and “Tools” respectively.

PACKAGER / MANUFACTURER shall submit the following shipping documents (Dispatch Dossier), containing at least:

a) Inspection Release Certificate (IRC), issued by HULL SUPPLIER or its representatives, which releases the equipment/material shipment;

b) Certificate of the Classification Society, if applicable;

c) Emergency card or "Material Safety Data Sheet", if applicable;

d) Requirements and instructions for lashing, protection, transportation, unloading and storage, whenever necessary;

e) General Arrangement and Data Sheet;

f) P&ID;

g) Packing List;

h) Lifting Scheme;

i) Packing, unpacking, preservation and storage procedures.

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9. DOCUMENTATION

PACKAGER / MANUFACTURER design drawings shall show all necessary dimensions and details required for interface information and installation. Clearances for maintenance shall be shown. Drawings and documents shall be clear and completely legible with all text in the English language.

Instruction manuals for operation and maintenance of the equipment/PACKAGE shall be provided in Portuguese language.

PACKAGER shall submit all major drawings and drawings containing interface information for HULL SUPPLIER approval according to the PACKAGER list. . Drawings are only accepted when signed by PACKAGER as checked and approved.All revised editions of drawings or documents shall show the revisions clearly marked up, the issue date and PACKAGER's checked and approved signatures.

PACKAGER / MANUFACTURER shall produce a weight/center of gravity data sheet, which shall reflect the individual component, and assembly weights, and also all necessary center of gravity details, considering with or without operational fluids.

PACKAGER shall send in advance all recommendations for PACKAGE installation, maintenance and commissioning.

PACKAGER shall issue a PACKAGE / Equipment Data Book to be delivered to HULL SUPPLIER for approval. Data Book minimum content shall be as the following:

1) certified drawings, data sheets, technical specifications, performance curves and calculation memorandum;
2) construction, maintenance and operating manuals, instructions for preservation and commissioning, and all catalogs, including of the sub-suppliers;
3) all certificates of materials and equipment, certificates of electrical cables and equipment to hazardous areas, all tests, destructive and non-destructive examinations, test reports (including FAT), certificates and reports of classification society, procedures for welding qualifications and welding processes.
4) the documentation requested by Brazilian law NR-13, subdivided for equipment (if applicable);
5) the documentation requested by Brazilian law NR-10, subdivided for equipment (if applicable);

Data Book delivery standard and conditions including number of parts and sections, number of printed and electronic copies will be further defined by OWNER.

10. TRAINING

PACKAGER shall provide training to qualify OWNER technicians to operate and maintain (install, dismantle, replace parts, make adjustments, etc.) each equipment. The training shall encompass all items to its understanding. The training shall be performed at FPSO construction yard and/or aboard the FPSO, after completion of the FAT and prior to OWNER approval of the FPSO Acceptance Term. PACKAGER shall provide all documentation and materials required for the training program (including the latest revision of the as built documentation, brochures, booklets, material for presentations, transparencies, etc.). Each individual equipment training program shall encompass all operation and maintenance aspects. All trainees will be operation and maintenance professionals. The participants shall be awarded certificates after the completion of the training course. The maintenance training shall be held to twelve (12) maintenance technicians and the operation training to twelve (12) operators , in Brazilian Portuguese and shall be performed using identical equipment as supplied or using the same equipment supplied at FAT phase. PACKAGER shall take full responsibility for the professionals teaching the training course, including for their transportation and lodging.

PACKAGER shall submit for OWNER approval the detailed training program. The training program shall cover, at least, the following items:

a) Complete description of the equipment;

b) Technical and operational characteristics;

c) Operating principles;

d) Operating cautions;

e) Aspects of construction;

f) Operating procedures and routines;

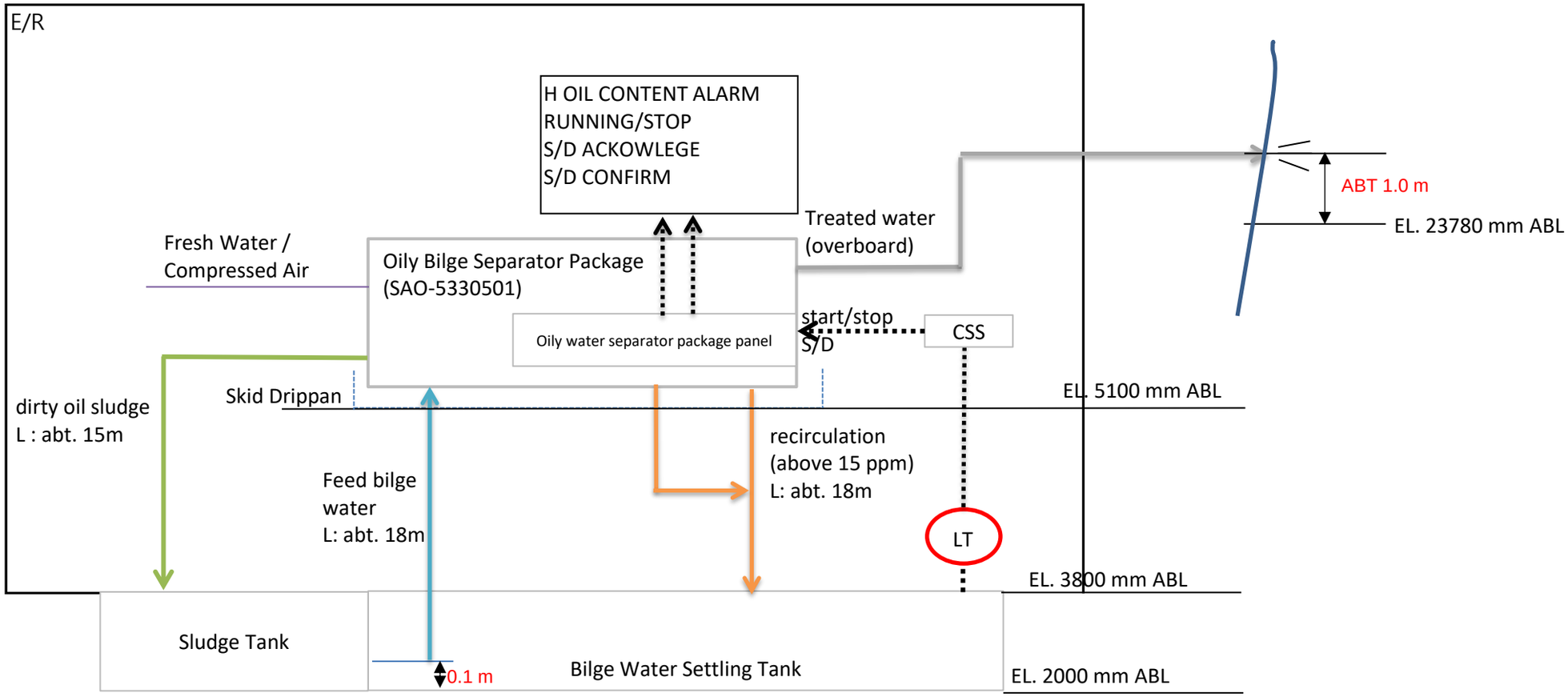
g) Identification of operational problems and possible causes (trouble-shooting);

h) Preventive maintenance routines;

i) Signaling and warning devices;

j) Protection and adjustment;

k) System drawings and diagrams

Appendix A

Schematic Diagram of Oily Water separator package interface